Unit: 6 Unit 7: Factoring
Topic: Difference of Squares

Name: ______________
Date: __________

Advanced (Habanero)

Q1. Factor the expression $x^2 - 4$
(A) $(x+2)(x-2)$
(B) $x(x-4)$
(C) $x(x+4)$
(D) $x^2 - 2x + 4$
(E) $x^2 + 2x - 4$

Q2. Factor the expression $25a^2 - 9b^2$
(A) $(5a-3b)(5a+3b)$
(B) $(5a+3b)(5a-3b)$
(C) $(25a+9b)(a-b)$
(D) $(25a-9b)(a+b)$
(E) $(5a+3b)^2$

Q3. Simplify $(x^2-4)(x^2-9)$ using difference of squares
(A) $(x-2)(x+2)(x-3)(x+3)$
(B) $(x-2)(x+2)(x^2-9)$
(C) $(x^2-4)(x-3)(x+3)$
(D) $(x^2-13)(x^2-3)$
(E) $(x^2-12)(x^2-1)$

Q4. Factor $16x^2 - y^2$
(A) $(4x-y)(4x+y)$
(B) $(16x+y)(x-y)$
(C) $(16x-y)(x+y)$
(D) $(4x+y)^2$
(E) $16(x^2-y^2)$

Q5. Factor $49x^2 - 16y^2$
(A) $(7x+4y)(7x-4y)$
(B) $(7x-4y)(7x+4y)$
(C) $(49x+16y^2)(x-y)$
(D) $(49x-16y^2)(x+y)$
(E) $49(x^2-y^2)$

Q6. Simplify $(x^2-9)(x^2-16)$ using difference of squares
(A) $(x-3)(x+3)(x-4)(x+4)$
(B) $(x-3)(x+3)(x^2-16)$
(C) $(x^2-9)(x-4)(x+4)$
(D) $(x^2-25)(x^2-4)$
(E) $(x^2-24)(x^2-5)$

Q7. Factor $64a^2 - b^2$
(A) $(8a-b)(8a+b)$
(B) $(8a+b)(8a-b)$
(C) $(64a+b^2)(a-b)$
(D) $(64a-b^2)(a+b)$
(E) $64(a^2-b^2)$

Q8. Factor $81x^2 - 4y^2$
(A) $(9x-2y)(9x+2y)$
(B) $(9x+2y)(9x-2y)$
(C) $(81x+4y^2)(x-y)$
(D) $(81x-4y^2)(x+y)$
(E) $81(x^2-y^2)$

Open Ended Questions (FRQ)

Q9. Factor the expression $36a^2 - b^2$ and simplify $(x^2-4)(x^2-9)$ using difference of squares

Q10. Prove that $(x^2-a^2)(x^2-b^2)$ can be factored as $(x-a)(x+a)(x-b)(x+b)$ and provide an example

Chef's Tips

- Encourage students to use the formula $a^2 - b^2 = (a+b)(a-b)$ to factor expressions
- Remind students to check their work and ensure that their factors are correct
- Suggest that students use real-world examples to help illustrate the concept of difference of squares